Lab 3 · A Look Under the Hood: the Command Line

Day 1 · 15:45–16:45 · 60 min · Pairs (swap driver/navigator at Part C) Objective: LO3 — Understand what Galaxy does behind the buttons, and take a first step towards independent analysis

Why bother? Every Galaxy tool you clicked today is a **command-line program** running on a computer somewhere. You don't need to become a programmer, but knowing a dozen commands means you can look inside huge files, check data from a sequencing provider, and follow online tutorials. This is the most **durable** skill in this course.

Golden rules: commands are **case-sensitive** · spaces matter · press **Enter** to run · press ↑ to repeat the last command · **Tab** auto-completes file names · **Ctrl + C** stops a command that is running.

Part A — Open a terminal inside Galaxy (10 min)

1. On **usegalaxy.eu**, search the tools for `Interactive JupyterLab Notebook` and click **Run tool** (you don't need to change any settings)
2. Wait 1–3 minutes. Go to **User → Active InteractiveTools** and click the link when it appears
3. In JupyterLab: **File → New → Terminal**
4. You will see a prompt ending in `$`. This is where you type.

If interactive tools are down: use your own laptop's **Terminal** (macOS/Linux) or **Git Bash** (Windows). The instructor has offline copies of the files.

Part B — Where am I? Moving around (15 min)

Type each command and press Enter. Write down what happens.

Command	What it does	Try it
`pwd`	**p**rint **w**orking **d**irectory: where am I?	`pwd`
`ls`	**ls**t files here	`ls`
`mkdir`	**m**a**k**e a **dir**ectory (folder)	`mkdir ssi_course`
`cd`	**c**hange **d**irectory	`cd ssi_course`
`cd ..`	go **up** one level	`cd ..` then `pwd`

Now go back into your new folder and stay there:

```
cd ssi_course
pwd
```

✍️ **Check yourself B:** Draw a little "tree" of the folders you have moved through.

Part C — Get real data and look inside (20 min) · *swap driver and navigator*

C1. Download sequencing reads

This is **the same Illumina read 1 file** you used in Lab 2:

```
wget https://zenodo.org/record/10669812/files/DRR187559_1.fastqsanger.bz2
ls -lh
```

`ls -lh` shows file sizes in **h**uman-readable units (K, M, G).

C2. Peek at the start of a compressed file

The file is compressed (`.bz2`). `bzcat` decompresses it **on the fly** and `head` shows only the first lines:

```
bzcat DRR187559_1.fastqsanger.bz2 | head -8
```

The `|` symbol is a **pipe**: it sends the output of one command into the next.

✍️ **Check yourself C2:** How many reads did you just see? Point to the quality line of the first read.

C3. Count the reads

FASTQ has 4 lines per read, so: **count lines, then divide by 4.**

```
bzcat DRR187559_1.fastqsanger.bz2 | wc -l
```

*(`wc -l` = **w**ord **c**ount, **l**ines. It may take about 20 seconds.)*

✍️ **Check yourself C3:**

1. How many reads are in this file? Does it match the "Total sequences" number in your Lab 2 FastQC report?
2. ⚠️ **Trap:** someone suggests counting reads with `grep -c "^@"` (count lines starting with `@`). Why could that give the **wrong** answer? *(Hint: look at the quality symbols table from Lab 1. Which Q score is `@`?)*

C4. Search inside the file

`grep` finds lines that contain a pattern. How many reads contain a run of 10 A's?

```
bzcat DRR187559_1.fastqsanger.bz2 | grep -c "AAAAAAAAAA"
```

Part D — From reads to genomes (10 min)

Download the assembled genome of the same isolate (we'll use it tomorrow):

```
wget https://zenodo.org/record/10572227/files/DRR187559_contigs.fasta
head -3 DRR187559_contigs.fasta
grep -c ">" DRR187559_contigs.fasta
```

✍️ Check yourself D:

1. What does `grep -c ">"` count, and why?
2. Compare it with the **# contigs** in your QUAST report from Lab 2.

Part E — Reflection (5 min)

You did this in Galaxy…	…the command line equivalent
Upload → Paste URL	`wget URL`
👁️ eye icon	`head`, `less`
Line/Word/Character count	`wc -l`
(not used today; the Galaxy tool is "Select lines that match an expression") Searching inside a file	`grep`
History	`history` (try it!)

When you finish: go back to Galaxy → **User** → **Active InteractiveTools** → **Stop** your JupyterLab, to free up resources.

Command cheat sheet (keep this)

```
pwd                 where am I?
ls  /  ls -lh       list files / with sizes
cd folder / cd ..   go into a folder / go up
mkdir name          make a folder
head -n 8 file      first 8 lines
wc -l file          count lines
grep "text" file    find lines containing text   (-c = count them)
bzcat / zcat file   view a .bz2 / .gz file without unzipping
cmd1 | cmd2         pipe: send output of cmd1 into cmd2
wget URL            download a file
history             list commands you have run
```

Want more? Work through the Software Carpentry **Unix Shell** lesson (about 4 hours, self-paced): https://swcarpentry.github.io/shell-novice/. You can do it in this same

JupyterLab terminal; the GTN version is at training.galaxyproject.org → Data Science → "CLI basics".

Adapted from The Carpentries "The Unix Shell" and the Galaxy Training Network "CLI basics" tutorial (CC-BY 4.0).